

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 Total Marks: 50

Annexure A

CODING CHALLENGE

Duration: 2hrs1.

Code of Conduct:

I K.Neeha(192425245)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.Neeha

1.Predict the Reinforcement Learning (RL) Framework

Definition

Reinforcement Learning (RL) is a machine learning approach in which an agent learns to make optimal decisions by interacting with an environment. The agent learns through trial and error by receiving rewards or penalties for its actions.

Agent–Environment Interaction

1. The agent observes the current state (S) of the environment.
2. It selects an action (A) based on its policy.
3. The environment responds by moving to a new state (S').
4. The agent receives a reward (R) that indicates whether the action was good or bad.
5. This process repeats until the task is completed.

Interaction Flow:

Environment → State (S)



Agent



Action (A)



Environment



Next State (S') + Reward (R)

Importance of States, Actions, and Rewards

- State (S): Represents the current situation of the environment.
- Action (A): Decision taken by the agent in a particular state.
- Reward (R): Feedback received after performing an action. Positive rewards encourage good actions, while negative rewards discourage poor actions.

Cumulative Reward

The objective of RL is not to maximize immediate reward, but to maximize the total (cumulative) reward over time.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

where:

- G_t = cumulative reward (return)
- R = reward
- γ = discount factor ($0 \leq \gamma \leq 1$)

Influence on Decision-Making

- The agent selects actions that provide the highest long-term cumulative reward.
- It may sacrifice an immediate reward if it leads to a greater future reward.
- This helps the agent learn an optimal policy for decision-making.

Example

A robot learns to reach a charging station:

- State: Robot's current location.
- Action: Move left, right, forward, or backward.
- Reward: +10 for reaching the charger, -1 for hitting an obstacle.
- The robot learns the shortest and safest path by maximizing cumulative reward.

Conclusion

The RL framework enables an agent to learn through interaction with the environment. By using states, actions, rewards, and cumulative reward, the agent gradually develops an optimal strategy that maximizes long-term performance.

2. Estimate how decision-making problems can be modeled as a Markov Decision Process (MDP).

Definition

A Markov Decision Process (MDP) is a mathematical framework used in Reinforcement Learning to model sequential decision-making problems where an agent interacts with an environment to maximize cumulative reward.

An MDP is represented as:

$MDP = (S, A, P, R, \gamma)$

where:

- S = Set of States
- A = Set of Actions
- P = Transition Probability
- R = Reward Function
- γ = Discount Factor

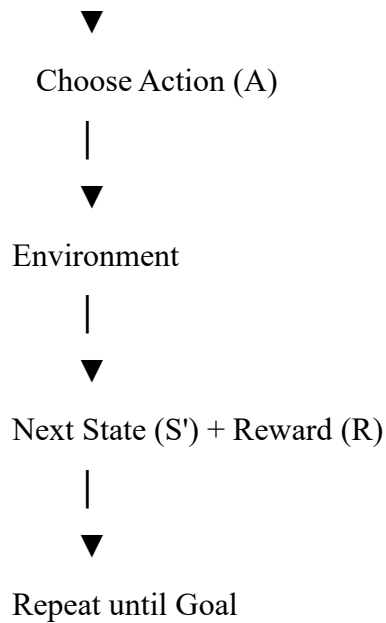
Components of MDP

Component	Description
State Space (S)	Represents all possible states of the environment. Example: Robot's current position.
Action Space (A)	Represents all possible actions the agent can perform. Example: Move Left, Right, Up, Down.
Transition Probability (P)	Probability of moving from one state to another after taking an action. It is represented as $P(s' s, a)$.
Reward Function (R)	Gives immediate feedback after performing an action. Positive reward encourages good actions, while negative reward discourages poor actions.
Discount Factor (γ)	A value between 0 and 1 that determines the importance of future rewards. Higher γ gives more importance to future rewards.

MDP Decision Process

Current State (S)

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Example

A robot navigating a maze:

- State (S): Robot's current location.
 - Action (A): Move Up, Down, Left, Right.
 - Transition Probability (P): 90% chance of moving correctly, 10% chance of slipping.
 - Reward (R): +10 for reaching the goal, -1 for hitting a wall.
 - Discount Factor (γ): 0.9 to prioritize long-term success.
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Conclusion

An MDP models decision-making by defining states, actions, transition probabilities, rewards, and the discount factor. It enables the agent to choose actions that maximize the expected cumulative reward, making it the foundation of Reinforcement Learning.

3. Discuss the role of policy and value functions in Reinforcement Learning (RL), and examine how state-value and action-value functions contribute to evaluating long-term benefits of actions using the Bellman Equation.

Policy in Reinforcement Learning

A policy (π) is a strategy that tells the agent which action to take in a given state. It maps states to actions and guides the agent toward maximizing long-term rewards.

Policy:

[
 $\pi(s)=a$
]

where:

- s = current state
 - a = action selected by the agent
-

Value Functions

Value functions estimate the expected cumulative reward that an agent can obtain in the future.

1. State-Value Function ($V(s)$)

- Measures the expected return starting from a state s and following policy π .
- It evaluates how good a state is.

$$[V_{\pi}(s) = E_{\pi}[G_t | S_t = s]]$$

2. Action-Value Function ($Q(s,a)$)

- Measures the expected return after taking action a in state s and then following policy π .
- It evaluates how good a specific action is.

$$[Q_{\pi}(s,a) = E_{\pi}[G_t | S_t = s, A_t = a]]$$

Bellman Equation

The Bellman Equation breaks a complex decision problem into immediate reward + discounted future reward, making it easier to compute optimal values.

State-Value Bellman Equation

$$[V(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) \left[R(s,a,s') + \gamma V(s') \right]]$$

Action-Value Bellman Equation

$$[Q(s,a) = \sum_{s'} P(s'|s,a) \left[R(s,a,s') + \gamma \sum_{a'} \pi(a'|s) Q(s',a') \right]]$$

where:

- $P(s'|s,a)$ = Transition probability
 - $R(s,a,s')$ = Immediate reward
 - γ = Discount factor
 - $V(s')$ = Value of next state
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Contribution to Long-Term Decision Making

- Policy decides which action to perform.
 - State-value function evaluates the quality of a state.
 - Action-value function evaluates the quality of each action in a state.
 - Bellman Equation updates these values recursively, helping the agent choose actions that maximize long-term cumulative rewards instead of immediate rewards.
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Example

A robot navigating to a charging station:

- State: Robot's current position.
 - Policy: Move toward the charger.
 - State Value: Higher near the charging station.
 - Action Value: Moving toward the charger has a higher Q-value than moving away.
 - Using the Bellman Equation, the robot updates these values after every step and learns the optimal path.
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Conclusion

Policy and value functions are fundamental to RL. The policy determines the agent's behavior, while state-value and action-value functions evaluate future rewards. The Bellman Equation enables efficient learning by combining immediate rewards with future expected rewards, allowing the agent to make optimal long-term decisions.

4. Justify the exploration–exploitation trade-off in Reinforcement Learning (RL) and compare strategies such as ϵ -greedy and softmax approaches in balancing learning efficiency.)

Exploration–Exploitation Trade-off

In Reinforcement Learning, an agent must balance exploration and exploitation to learn an optimal policy.

- Exploration: Trying new or less-visited actions to discover better rewards.
- Exploitation: Choosing the action with the highest known reward based on current knowledge.

A good balance between the two helps the agent learn efficiently and achieve maximum cumulative reward.

Need for the Trade-off

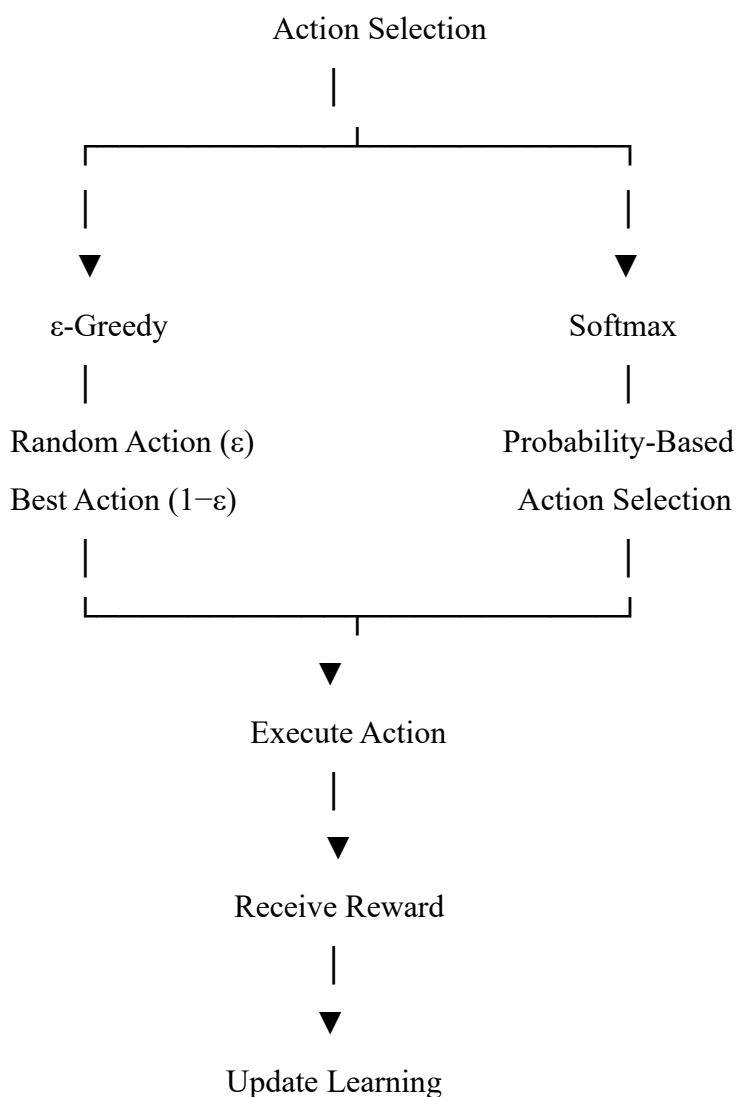
- Too much exploration: The agent spends time testing many actions and learns slowly.
- Too much exploitation: The agent may miss better actions and become stuck in a suboptimal policy.
- Therefore, balancing both strategies improves learning performance.

Comparison of ϵ -Greedy and Softmax Strategies

Example

Consider a robot choosing among four paths to reach a destination:

- ϵ -Greedy: Chooses the best-known path 90% of the time and a random path 10% of the time.
 - Softmax: Chooses all paths with probabilities based on their estimated rewards, giving better paths higher chances while still exploring others.
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Conclusion

The exploration–exploitation trade-off is essential in RL because it balances discovering new actions with using the best-known actions. ϵ -Greedy is simple and effective, while Softmax offers more intelligent exploration by selecting actions according to their estimated values, leading to improved learning efficiency in many environments.

5. Explain how reward shaping can accelerate learning in Reinforcement Learning (RL) and illustrate its impact on preventing suboptimal policies. (5 Marks)

Definition

Reward shaping is a technique in Reinforcement Learning where additional intermediate rewards are provided to guide the agent toward the desired goal. It helps the agent learn faster by giving more informative feedback instead of rewarding only the final outcome.

How Reward Shaping Accelerates Learning

- Provides frequent feedback during learning.
 - Encourages the agent to take actions that lead toward the goal.
 - Reduces the time required to discover successful strategies.
 - Improves learning efficiency by reducing unnecessary exploration.
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Impact on Preventing Suboptimal Policies

A suboptimal policy is a strategy that gives a satisfactory result but is not the best possible solution.

Reward shaping helps prevent suboptimal policies by:

- Encouraging actions that lead to higher long-term rewards.
- Discouraging inefficient or harmful actions with negative rewards.
- Guiding the agent away from local optima toward the optimal policy.

Example

A robot must reach a charging station.

- Without Reward Shaping:
 - Reward = +100 only when the robot reaches the charging station.
 - The robot may wander randomly for a long time before learning.
- With Reward Shaping:
 - +5 for moving closer to the charging station.
 - -2 for moving away.
 - +100 for reaching the goal.

The robot learns the correct path much faster and avoids unnecessary movements.

Advantages

- Speeds up the learning process.
- Improves convergence to the optimal policy.
- Reduces exploration time.

- Increases overall learning efficiency.

Conclusion

Reward shaping improves Reinforcement Learning by providing intermediate rewards that guide the agent toward the goal. It accelerates learning, reduces unnecessary exploration, and helps prevent suboptimal policies, enabling the agent to learn an optimal strategy more efficiently.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

		behavior throughout			
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately